Utilizing Solar Thermal Energy for Hostel Cooking

Aayush M Anand (21d170001), Brahadish Balaji D (23m1341), Prasad Madhukar Pawar (23d0640),
Vishnu T (22n0413) Group 5, CP1

Abstract

The Course Project helps us to explore the feasibility and implementation of utilizing solar thermal energy for cooking in hostel-17 of IIT Bombay accommodating approximately 1300 students. The proposed solution involves the deployment of Scheffler dish solar collectors to harness and concentrate sunlight for cooking purposes. The project assesses the energy requirements for cooking in the hostel, considering the needs and demands of the students over different meals of the day. The investigation involves analyzing the solar irradiance in the geographical location of the hostel using the weather file, determining the optimal number and size of Scheffler dishes required, and evaluating the potential energy savings. Additionally, considerations are made for energy storage systems to ensure continuous cooking capabilities during periods of low solar availability and for off-sun hours.

Keywords: Hostel Cooking, Scheffler Dish, Thermal Energy Storage, Heat transfer fluid, Concentration ratio

1. Introduction

India envisions an ambitious target of net zero by 2070. Our course project has aimed at the cooking sector and assessing various alternatives and pathways to adopt to achieve the target. India has an active gas consumer base of 32 crores and demand is largely import-dependent. Renewable energy-driven cooking will help in energy independence and reduce carbon emissions to almost zero. Further, it is cost-effective in the long run.

Our hostel in IIT Bombay is a residence to 1300 students consuming over 5900 MJ per day only for cooking consuming 9 LPG cylinders of 14.2 kg. This study assesses an alternative to solar thermal cooking using a Scheffler dish concentrator. Scheffler collectors use a large, flat, fixed mirror to reflect sunlight onto a fixed focal line or point. Unlike the parabolic shape, the mirror in Scheffler collectors is flat, and it is inclined at a specific angle to concentrate sunlight, which is mainly used for cooking. We will be using a mechanical tracking system to keep the mirror oriented towards the sun, ensuring continuous concentration of sunlight on the fixed focal line. Scheffler collectors are generally simpler in design and may be more cost-effective compared to parabolic dishes.

2. Application

Application Type - Residential.

Food consumption pattern over a day and its breakdown:

Energy Calculations per day:

1) Gas cylinders (14.2kg) - 9 Nos .

LPG has a calorific value of 46.1 MJ/kg.

$$Total\ energy\ =\ mass\ x\ calorific\ value \tag{1}$$

14.2 cylinder has an energy content of 654.6 MJ of heat.

We have a total of 9 Cylinders which corresponds to total energy used per day = 5900MJ

H17	1300 PEOPLE
Total LPG Gas per day	5900 MJ
Consumption distribution:	
Milk steamer	300 litres
Rice	240 kg
Roti	240 kg
Curry & Dal	90 kg
Vegetable side dish	90 kg
Non veg	30 kg
Oil frying	9 litres
Hot water	100 litres

Table 1: Energy consumption distribution

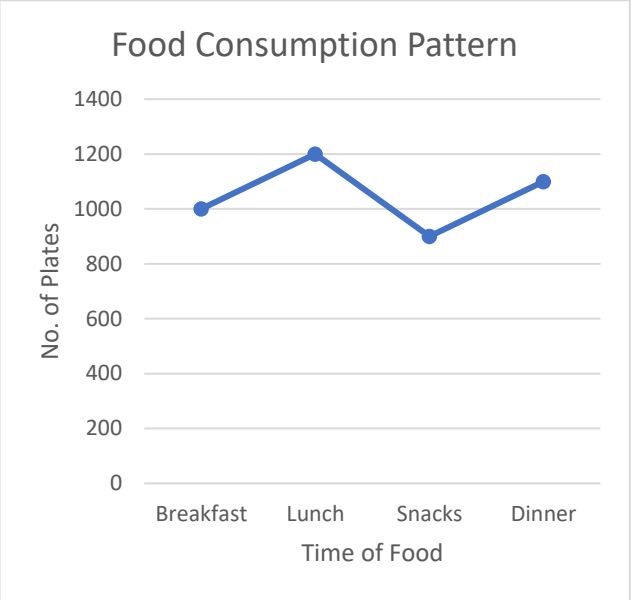


Fig 1: Energy Demand Pattern over a day

3. Solar collector system

Scheffler dish is a small lateral section of a paraboloid, which concentrates the sun's radiation over a fixed focus, with automatic single-axis tracking.The system uses steam, operating at 150–170^{0}C temperature at 3 kg/cm^2 pressure.

Working Principle

The Scheffler dish is made of several flat-shaped mirrors or reflective films that are mounted on a structural frame. The dish rotates about a north-south axis parallel to the earth's axis to track the sun's movement. The axis of rotation passes through the center of gravity of the reflector and that's how the reflector always maintains its gravitational equilibrium. The Scheffler reflector also performs a change in inclination angle while staying directed to the sun, in order to obtain a sharp focal point. Focus is fixed at the distance of the focal length of the paraboloid along the axis of the paraboloid. The receiver at a fixed location captures the concentrated heat and transfers it to water/thermic fluid to generate hot water/hot thermic fluid or high-pressure steam.

Subsystem

The receiving subsystem collects solar radiation in the solar field from parabola-shaped reflecting collectors and transfers it to working fluids such as water or oil. The tracking subsystem acts as a support to the receiving subsystem. It maximizes the solar energy captured by moving solar collectors towards the sun for a complete day. The circulation subsystem carries fluid and transfers the heat received by it to the end-use application.

Components

Collector - The collector of Scheffler Dish is an assembly of flat-shaped solar-grade glass mirrors or Aluminium mirror reflectors arranged on a structural steel framework. The mirrors are arranged to give a paraboloidal shape and reflect the incident solar radiation to a common point, i.e. the receiver.

Receiver - The receiver of the Scheffler dish is placed at the focus of the dish so as to capture the incident solar radiation and transfer it to the thermal medium used in the system. The critical component is generally a dome-type absorber fixed onto the structural steel frame. The receiver works on the thermo siphon principle in which cold water sinks down from the header and hot water/steam moves up to the header.

Header Pipe
• Storage tank contains water in the lower half and steam in the upper half
• Works as a buffer system, i.e. it collects and stores steam generated by the Scheffler system, which can be utilized as per the requirement

4. Conclusion

In this project, we try to explore the feasibility and potential benefits of implementing a Scheffler-based solar collector for cooking applications in Hostel 17. Although the actual deployment and observations requires very detailed studies, our project intends to shed light on several aspects.

1. Feasibility Analysis
2. Anticipated Benefits
3. Environmental Impact
4. Community Engagement

Moving forward, the outcome of the project can be used as a quantitative assessment for implementing the project in real life.
In conclusion, while the actual deployment and observations requires further technical and on ground studies, our project intends to shed light on the potential of a Scheffler-based solar collector as a sustainable and eco-friendly solution for cooking in Hostel 17.

Nomenclature

Symbols

A	Area, m^2
P	Pressure, bar
T	Temperature, K
U	Overall heat transfer coefficient, W m^{-2} K^{-1}
MJ	MegaJoule
kg	kilogram

References

[1]. Scheffler Dish - Taylormade Renewables Ltd. Date Accessed January 21, 2024 https://trlindia.com/wp-content/uploads/2020/12/TRL-Parabolic-dish-based-Solar-System-Operations-Maintenance-Manual-1.pdf
[2]. Material and Component Specifications Fixed Focus Automatically Tracked Elliptical Dish (Scheffler) Date Accessed January 21, 2024 https://itpower.co.in/wp-content/uploads/2016/01/Final-Booklet-2-Scheffler-Dish.pdf
[3]. https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1945472
[4]. https://www.tribuneindia.com/news/archive/features/a-peep-into-piped-comfort-660183
[5]. Comparative Evaluation of Parabolic Collector and Scheffler Reflector For Solar Cooking,Swati Bhasme